Semisupervised Hyperspectral Image Classification Using a Probabilistic Pseudo-Label Generation Framework

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***Abstract*— Deep neural networks (DNNs) show impressive performance for hyperspectral image (HSI) classification when abundant labeled samples are available. The problem is that HSI sample annotation is extremely costly and the budget for this task is usually limited. To reduce the reliance on labeled samples, deep semisupervised learning (SSL), which jointly learns from labeled and unlabeled samples, has been introduced in the literature. However, learning robust and discriminative features from unlabeled data is a challenging task due to various noise effects and ambiguity of unlabeled samples. As a result, recent advances are constrained, mainly in the pretraining or warm-up stage. In this article, we propose a deep probabilistic framework to generate reliable pseudo-labels to explicitly learn discriminative features from unlabeled samples. The generated pseudo-labels of our proposed framework can be fed to various DNNs to improve their generalization capacity. Our proposed framework takes only ten labeled samples per class to represent the label set as an uncertainty-aware distribution (We use the Gaussian distribution to represent the uncertainty of the label set in the latent space.) in the latent space. The pseudo-labels are then generated for those unlabeled samples whose feature values match the distribution with high probability. By performing extensive experiments on four publicly available datasets, we show that our framework can generate reliable pseudo-labels to significantly improve the generalization capacity of several state-of-the-art DNNs. In addition, we introduce a new DNN for HSI classification that demonstrates outstanding accuracy results in comparison with its rivals.**

***Index Terms*— Deep neural network (DNN), hyperspectral image (HSI) classification, probabilistic embedding, pseudo-labeling, semisupervised learning (SSL).**

NOMENCLATURE

$\mathcal{I}^l, \mathcal{I}^u$	Labeled and unlabeled sets.
$\mathbf{I}^l, \mathbf{I}^u$	Samples of $\mathcal{I}^l$ and $\mathcal{I}^u$.
M, N	Total number of labeled and unlabeled samples.
L	Total number of classes.
C, D'	Dimension of HSI channels and encoded samples.
$\mathcal{I}'^l, \mathcal{I}'^u$	Encoded labeled and unlabeled sets.
$\mathbf{I}'^l, \mathbf{I}'^u$	Samples of $\mathcal{I}'^l$ and $\mathcal{I}'^u$.
$\mathbf{I}^{*l}$	Optimal latent vector of $\mathcal{I}^l$ samples.
$\mathcal{S}^l, \mathcal{S}^u$	Labeled and unlabeled similarity score sets.
s^l, s^u	Scalar values of $\mathcal{S}^l$ and $\mathcal{S}^u$.
$\mathbf{I}''^l$	Mean of $\mathcal{I}'^l$.
I''^l	Scalar value of $\mathbf{I}''^l$.
$\hat{\mu}^l_M, \hat{\sigma}^{l^2}_M$	Mean and variance of $\hat{p}^l$.
$\hat{\mathcal{I}}'^u$	Extracted set from $\mathcal{I}'^u$.
$\hat{\mathbf{I}}'^u$	Sample of $\hat{\mathcal{I}}'^u$.
R^l	Confidence interval range.
$\hat{\mathcal{I}}'^l, \hat{\mathcal{Y}}^l$	Pseudo-image and label sets.

I. INTRODUCTION

HYPERSPECTRAL imagery (HSI) is a 3-D form of images, capable of representing light intensity in various frequencies using hundreds of spectral bands. HSIs have been utilized in numerous applications, such as mineral exploration [1], light pollution [2], vegetation mortality [3], and stress detection [4]. Among these applications, supervised classification is an active research area that enables many downstream tasks to be performed with high accuracy [5]. Supervised classification has been addressed by a great number of methods, of which support vector machines (SVMs) [6], random forest (RF) [7], and maximum likelihood (ML) [8] are some of the conventional methods. However, robust and discriminative feature analyses have been challenging for these methods due to the nonlinearities and complexities of HSIs. To this end, advanced methods have applied deep neural networks (DNNs) to extract more powerful and discriminative features and obtain superior performance.

Various DNNs have been introduced in recent years in the field of HSI classification to overcome the limitations of traditional methods. Most of these DNNs are equipped with robust hierarchical feature extractors such as convolution neural networks (CNNs) or recurrent neural networks (RNNs) that extract more discriminative and abstract features [9]–[12]. Several DNNs were developed for HSI classification and demonstrated state-of-the-art performance [13]–[17]. However, DNNs typically need a considerable number of labeled samples to update their parameters. This is because DNNs' feature extractors contain thousands or even millions of parameters that need to be properly tuned. Since this process is

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labor-intensive and time-consuming, there are a limited number of labeled samples, presenting an ongoing obstacle to classification. To grapple with this issue, various approaches have been introduced in the literature to train DNNs with a limited number of training samples [18]–[22]. Among them, methods that exploit nontarget samples (e.g., unlabeled samples or samples from other data sources) to learn a DNN's parameters have received considerable attention [23]–[26]. These methods can be roughly categorized into four approaches: pretraining, active learning (AL), data augmentation (DA), and pseudo-label generation.

The pretraining approach includes the methods that use nontarget samples to preoptimize the parameters of DNNs' layers [21]. For instance, Yang *et al.* [22] use pretrained parameters for the low and mid-layers of their proposed CNN-based network. Lin *et al.* [27] propose a structural preserving strategy to use the pretrained parameters in an unsupervised manner. Zhou *et al.* [28] develop stacked autoencoder to pretrain the parameters in an unsupervised manner and use them in a classification network. A semisupervised method is proposed in [29], in which stacked autoencoders are separately applied to pretrain the classification network based on spatial and spectral features. Moreover, unsupervised HSI classification using pretrained parameters and knowledge transfer are proposed in [27]. Although these methods enhance the classification results by using pretrained networks, parameter adaptation and fine-tuning still remain the key challenges in this approach.

The second approach includes the methods that apply AL strategies to address labeled data scarcity for DNNs [23], [26]. In general, AL-based methods aim to learn a model from a small set of labeled samples and then represent the uncertainty of the unlabeled samples based on the learned model. Applying AL is not straightforward for DNNs due to the small label set selection and uncertainty representation; as a result, existing methods attempted to formulate AL using various techniques such as Bayesian-based representation [26], uncertainty estimation [30], or transfer learning [31]. For instance, Deng *et al.* [31] propose a deep active transfer learning framework in which an autoencoder is trained on a source domain and transferred to the target domain by taking advantage of AL. Despite the progress, the efficacy of AL for DNNs training is still under question [32].

The third approach is DA, which attempts to increase the number of labeled samples by constructing virtual samples. Early work in this approach aimed to use conventional methods such as affinity matrices to generate more labeled samples [33]. More advanced methods applied generative adversarial networks (GANs) [34] to construct more diverse virtual samples. In particular, by adversarial learning of two GAN components (i.e., generator and discriminator), the generator learned to produce perfect fake samples. For instance, Zhan *et al.* [35] propose a 1-D GAN to pretrain the network's parameters using generated fake data. Zhu *et al.* [15] apply a GAN to directly perform the classification task using original and fake samples. GANs have also been combined with other methods to improve classification outcomes. For instance, Zhong *et al.* [25] combine GAN with conditional random field (CRF) to improve the classification performance by imposing graph constraints on the probability prediction of the GAN's discriminator. Wang *et al.* [36] also combine DropBlock [37] with a GAN to generate fake samples that are spatially well-distributed. Nevertheless, while the GAN-based methods improve HSI classification by creating diverse fake samples, the generated samples cannot represent a complete distribution of real HSI samples [38].

The fourth approach includes the methods that propose DNN-based clustering-classification frameworks. In particular, these methods apply a clustering algorithm to generate labels for unlabeled samples ("pseudo-labels") and used them to train the DNNs. For instance, Wu and Prasad [39] propose a semisupervised method that uses a nonparametric clustering algorithm to generate pseudo-labels. Fang *et al.* [24] propose a joint clustering-classification method that iteratively generates pseudo-samples using a rank-order clustering algorithm [40]. Generating pseudo-labels is highly beneficial to guide the learning process by exploiting unlabeled samples. Nevertheless, pseudo-label methods constrain to a fine-tuning stage. Typically, this is because if the pseudo-labels are inaccurate, the classification model learns from incorrect samples. Learning from incorrect pseudo-labels leads to propagated errors and confirmation bias issues [41], [42].

In this article, we propose a pseudo-label generation framework based on probabilistic embedding. Our proposed framework, belonging to the fourth approach, mitigates the problem of inaccurate pseudo-label generation by measuring the uncertainty level of pseudo-labels and filtering out those with low confidence. The proposed framework offers a number of further advantages, such as a light and easy-to-implement framework, and high compatibility with various DNNs.

The contributions of this work can be summarized as follows.

1) We propose an uncertainty-aware probabilistic pseudo-label generation framework to effectively represent labeled samples as a distribution, which is taken into account to infer pseudo-labels for unlabeled samples.
2) Our probabilistic pseudo-label generation framework enables uncertainty estimation to filter out ambiguous samples. As a result, our generated pseudo-labels are confident enough that they can be directly applied during the training, without the need for any fine-tuning.
3) The probabilistic framework offers a lightweight pipeline to generate reliable pseudo-labels that can be fed to different types of supervised DNNs to remarkably improve their generalization performance when limited labeled data is available.
4) In addition to the probabilistic framework for pseudo-label generation task, a classification network (denoted as patch-related convolution long short-term memory (PRCLSTM) II) is developed to further enhance the HSI classification results by extracting discriminative spectral–spatial features.

The rest of this article is organized as follows. Section II describes related work in pseudo-labeling and uncertainty learning in the field of computer vision. Section III introduces

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our proposed framework and its components, elaborates on the strategy of encoding samples in the embedding space, and represents them in the form of distribution. Section IV demonstrates the experimental configurations and baseline DNNs for the comparison assessments. The classification performances of the DNNs are assessed and visualized based on four publicly available datasets. Finally, Section V provides a discussion and a conclusion regarding this work.

II. Related Work

In recent years, many approaches have been introduced to alleviate the problem of limited labeled samples for DNN training [43]–[49]. Among them, semisupervised methods that generate pseudo-labels for unlabeled samples have attracted considerable attention [24], [39], [50]. However, as mentioned earlier, incorrect pseudo-labels cause noise accumulation, that is, confirmation bias [41], [51]. In this regard, different approaches such as adversarial attacks [52], uncertainty weighting [50], graph consistency [53]–[56], and confidence regularization [29], [57] have been proposed to deal with noise accumulation. This study focuses on the uncertainty modeling of DNN's estimations [58], [59] to address the noise accumulation problem. The uncertainty modeling approaches attempt to address the uncertainty quantification of point-based instance embedding. The point-based embedding is a backbone of many state-of-the-art DNNs for HSI classification [9], [13], [60]–[63]. Typically, the procedure of point-based instances embedding boils down to representing labeled and unlabeled samples on a low-dimension latent space as vector points. The point-based embedding can be applied as a robust representation approach if the target latent space is well-defined using the high-quality labeled data [64], [65]. However, with the limited labeled data, mostly affected by noises and ambiguous features, the target latent space cannot guarantee to be well-defined. The uncertainty in the target latent space results in large shifts of the embedded points [58], [59], [64], [66] and leads to a false classification [67]. To tackle this problem, caused by the large degree of data uncertainty,[1] one approach is to apply the probabilistic embedding [64]. In comparison to the point-based embedding, which mostly performs uncertainty modeling using a metric function, the probabilistic embedding estimates the uncertainty of instances using the Gaussian-based distributions[2] [64], [72]. Recent studies have applied probabilistic embedding to estimate the latent space's uncertainties using the embedded features of the labeled samples [65], [67], [71]. Despite the effectiveness of the proposed approach for uncertainty modeling of the labeled samples, the unlabeled samples are out of concern for these recent studies. However, in this study, we leverage the HSI unlabeled samples and apply the probabilistic embedding to further estimate the uncertainty among labeled and unlabeled samples in the common latent space. The proposed approach leads to the utilization of probabilistic embedding for the task of semisupervised classification by generating pseudo-labels for the unlabeled samples.[3]

[1]The data uncertainty refers to the problem of ambiguity of the features (i.e., spectrum) and inherent noises [68].

[2]The Gaussian distribution has been widely used as a nonparametric prior distribution to be applied for uncertainty modeling in machine learning and computer vision [68]–[71].

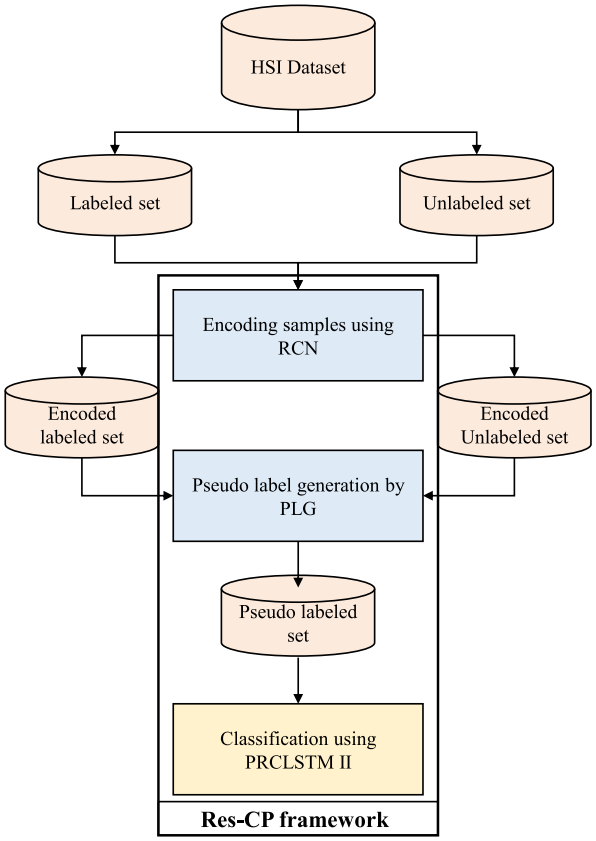


Fig. 1. High-level illustration of the proposed semisupervised probabilistic framework, called Res-CP.

III. Proposed Methodology

In order to address the problem of limited training samples for the classification task, we leverage the unlabeled samples to enhance the generalization capacity of DNNs. Our semisupervised framework, denoted as Res-CP,[4] uses a small number of training samples (i.e., ten samples per class) to generate a reasonable number of pseudo-labels for unlabeled samples. Unlabeled samples with their corresponding pseudo-labels (pseudo-samples) are then used to train DNNs for classification. Our Res-CP framework consists of three main components: residual convolution network (RCN), pseudo label generator (PLG), and PRCLSTM II. These are briefly described below.

1) RCN: A residual center-based network that encodes the labeled and unlabeled samples to a common embedding space.
2) PLG: A probabilistic pseudo-label generation method first estimates the uncertainty-aware distribution of the encoded samples and then takes the distribution into account to generate pseudo-labels. In particular, the PLG assumes the uncertainty of the encoded labeled samples of each class as a Gaussian distribution; the mean of

[3]The unlabeled samples, whose estimated uncertainty is lower than a defied confidence threshold, are selected for the task of pseudo label generation.

[4]https://github.com/majidseydgar/Res-CP

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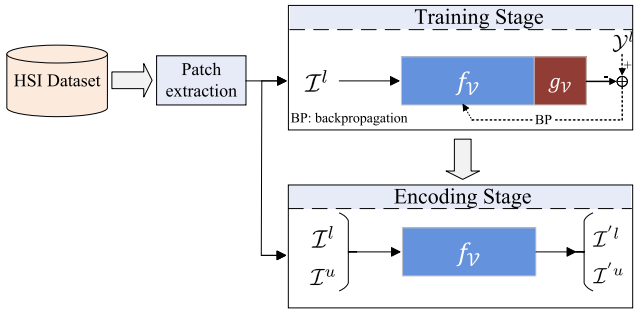


Fig. 2. Feature encoding pipeline using $f_\mathcal{V}$ and $g_\mathcal{V}$ components of RCN.

the estimated distribution represents the most confident embedded features of the labeled set and the variance indicates the uncertainty of the estimation. The distribution is then deployed as a reference to assess the confidence of the unlabeled samples belonging to the target domain and select the highly confident ones for the task of a pseudo-label generation.

3) PRCLSTM II: A modified version of the classification model PRCLSTM that is presented in [13]. Here, our developed model, PRCLSTM II, is equipped with a residual architecture to prevent gradient saturation and enhance the information flow throughout the network. The residual architecture enables PRCLSTM II to learn features from pseudo-samples that are more discriminative and improve the classification result.

In summary, the first two components of Res-CP perform pseudo-label generation and the third component performs classification to demonstrate the effectiveness of the generated pseudo-samples. It should be noted that the generated pseudo-labels can be used to train any DNNs to improve their generalization capacity, not only the PRCLSTM II. Fig. 1 shows the Res-CP framework and its components. We detail each of the Res-CP components in the following sections.

A. Feature Encoding Using RCN

In this section, we elaborate on the procedure for encoding labeled and unlabeled samples into a common embedding space. For all samples, we first extract a patch[5] from the HSI image with a size of $w \times w \times C$, where w represents the height and width of the patch, and C indicates the number of channels. Let $\mathcal{D} = \{\mathcal{I}^l, \mathcal{Y}^l\}_{l=1}^L$ denote the labeled set, where L denotes the total number of classes. Here, $\mathcal{I}^l = \{\mathbf{I}_m^l\}_{m=1}^M$ is a set of extracted patches, where $\mathbf{I}_m^l \in R^{w\times w\times C}$, and M is the total number of labeled samples in class l. Moreover, $\mathcal{Y}^l = \{y_m^l\}_{m=1}^M$ is a set of labels in class l. The labels and the image patches are connected via ground-truth ($\mathcal{G}$) matches. The unlabeled set is defined via $\mathcal{I}^u = \{\mathbf{I}_n^u\}_{n=1}^N$, where $\mathbf{I}_n^u \in R^{w\times w\times C}$, and N is the total number of unlabeled samples. The idea is to encode the samples into a common embedding space such that the posterior distribution represents the uncertainty of the input data. The first stage to achieve this aim is to properly optimize the embedding space with a deterministic encoding model. In order to achieve that, we apply a residual image encoder (RCN) consisting of two subcomponents $f_\mathcal{V}$ and $g_\mathcal{V}$. Fig. 2 illustrates the encoding procedure using $f_\mathcal{V}$ and $g_\mathcal{V}$. Note that $f_\mathcal{V}$ is a feature encoding component that uses spectral–spatial residual network (SSRN) [60] as its feature extraction backbone; it is equipped with 3-D CNN layers and residual connections [10]. In addition, $g_\mathcal{V}$ consists of two fully connected (FC) layers; it receives $f_\mathcal{V}$ output and computes the loss of the RCN parameters to update them using a joint loss ($\mathcal{L}_\mathcal{J}$). Fig. 3 shows the $f_\mathcal{V}$ and $g_\mathcal{V}$ layers in more detail. Here, $\mathcal{L}_\mathcal{J}$ is a center-softmax loss introduced in [75]. The center-softmax loss ($\mathcal{L}_\mathcal{J}$) consists of two losses: center and softmax. The center loss estimates the center of each class and forces the samples' features to match the corresponding center during the training stage [75]. In addition, the softmax loss encouraged features of each class in the sample to be more separable. Fig. 3 provides a visual example of each loss effect in the proposed RCN. The sample encoding procedure is summarized in Algorithm 1. Specifically, $f_\mathcal{V}$ and $g_\mathcal{V}$ are both applied at the first stage to update $f_\mathcal{V}$ parameters and learn the embedding space using the labeled samples. After that, $f_\mathcal{V}$ is applied to encode $\mathcal{I}^l$ and $\mathcal{I}^u$ into the learned embedding space. The outputs of Algorithm 1 are the encoded labeled and unlabeled samples, denoted as $\mathcal{I}'^l$ and $\mathcal{I}'^u$, respectively.

Algorithm 1 Encoding Procedure of $\mathcal{I}^l$ and $\mathcal{I}^u$ Into a Common Embedding Space Using RCN

Input: Given RCN consists of $f_\mathcal{V}$ and $g_\mathcal{V}$; $\mathcal{D} = (\mathcal{I}^l, \mathcal{Y}^l)$; $\mathcal{I}^u$; and the number of epochs, denoted as E_P.

Output: Encoded labeled and encoded unlabeled set, denoted as ($\mathcal{I}'^l$) and ($\mathcal{I}'^u$), respectively.

```
i ⟵ 0
while i < E_P do
    Enter D samples to f_V to compute L_J using g_V
    Backpropagate L_J loss to update f_V, and g_V parameters
    i ⟵ i + 1
end while
for l = 1, ..., L do
    for m = 1, ..., M do
        Generate I'^l = {g_V(I_m^l)}_{m=1}^M
    end for
    for n = 1, ..., N do
        Generate I'^u = {g_V(I_n^u)}_{n=1}^N
    end for
end for
```

B. Pseudo-Label Generation Using PLG

In this section, we describe the procedure of distribution estimation for labeled samples in the embedding (i.e., latent) space to match the unlabeled samples for a pseudo-sample generation. As illustrated in Fig. 4, we estimate a Gaussian distribution in the embedding space as a reference for the label set. Next, by encoding the unlabeled samples into the common embedding space, we estimate their similarities to the reference distribution. The samples with a high probability of a

[5]The patch extraction is a common and widely used practice in various spectral–spatial state-of-the-art supervised and semisupervised HSI classification methods [13], [17], [57], [60], [73], [74].

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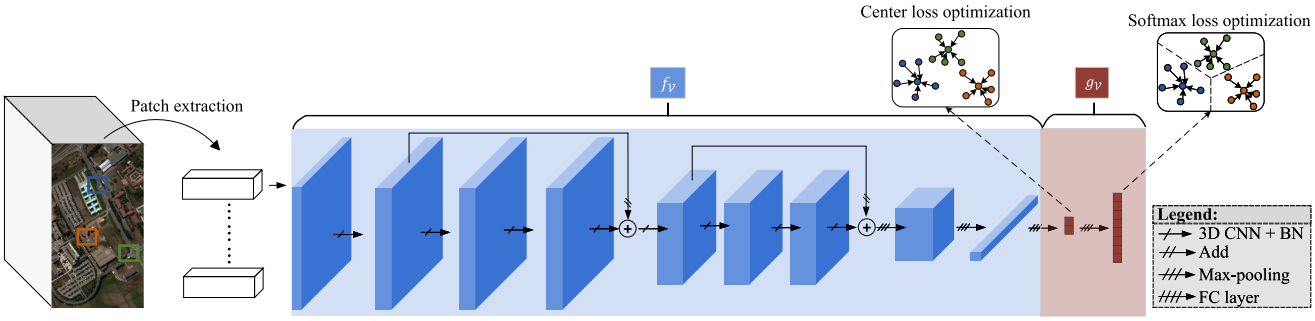


Fig. 3. Illustration of $f_{\mathcal{V}}$ and $g_{\mathcal{V}}$ layers in detail.

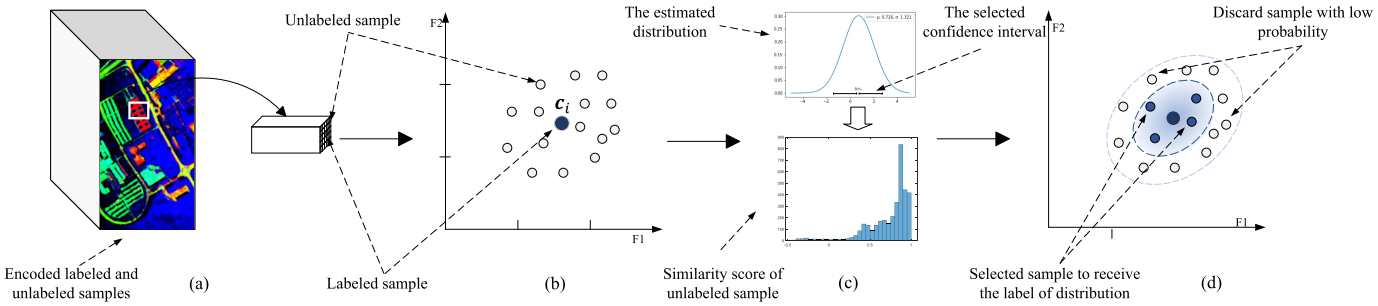


Fig. 4. Illustration of pseudo-sample generation procedure of the PLG algorithm. (a) Nearby unlabeled samples extraction. (b) 2-D samples illustration in the feature space. (c) Similarity score estimation. (d) Filter out low confidence samples and pseudo-label generation.

match are selected for the pseudo-label generation task, while the low probability samples are filtered out. This procedure consists of two core components: distribution estimation and unlabeled sample matching.

1) Distribution Estimation: After generating the encoded features of labeled samples (i.e., $\mathcal{I}'^l$), the aim is to construct a probabilistic representation to determine the distribution of the latent codes in the embedding space. The intuition behind estimating a distribution is that representing the encoded patches by a sole latent code is unrealistic due to the existence of noisy and low-quality samples. The probabilistic embedding enables us to represent the latent code in a form of distribution. Let us consider $(\mathbf{I}'^l_1, \ldots, \mathbf{I}'^l_M)$ to be the encoded samples in a certain class, l, and $\mathbf{I}^{*l} \in \mathrm{R}^{1\times D'}$ to be the optimal latent code of the image patches. The assumption is that the encoded samples can be represented by a Gaussian distribution. By assuming that the image patches, that is, $(\mathbf{I}^l_1, \ldots, \mathbf{I}^l_M)$, are conditionally independent [58], [59], [76], the posterior distribution, $p(\mathbf{I}^{*l} \mid \mathbf{I}^l_1, \ldots, \mathbf{I}^l_M)$, can be expressed as follows:

$$\begin{aligned} &p(\mathbf{I}^{*l} \mid \mathbf{I}^l_1, \ldots, \mathbf{I}^l_M) \\ &= \frac{p(\mathbf{I}^l_1, \ldots, \mathbf{I}^l_M \mid \mathbf{I}^{*l})\, p(\mathbf{I}^{*l})}{p(\mathbf{I}^l_1, \ldots, \mathbf{I}^l_M)} \\ &= \frac{p(\mathbf{I}^l_1, \ldots, \mathbf{I}^l_{M-1} \mid \mathbf{I}^{*l})\, p(\mathbf{I}^l_M \mid \mathbf{I}^{*l})\, p(\mathbf{I}^{*l})}{p(\mathbf{I}^l_1, \ldots, \mathbf{I}^l_M)} \\ &= \alpha \frac{p(\mathbf{I}^l_1, \ldots, \mathbf{I}^l_{M-1} \mid \mathbf{I}^{*l})\, p(\mathbf{I}^l_M \mid \mathbf{I}^{*l})\, p(\mathbf{I}^{*l})}{p(\mathbf{I}^l_1, \ldots, \mathbf{I}^l_{M-1})\, p(\mathbf{I}^l_M)} \\ &= \alpha \frac{p(\mathbf{I}^l_1, \ldots, \mathbf{I}^l_{M-1}, \mathbf{I}^{*l})\, p(\mathbf{I}^l_M, \mathbf{I}^{*l})}{p(\mathbf{I}^l_1, \ldots, \mathbf{I}^l_{M-1})\, p(\mathbf{I}^l_M)\, p(\mathbf{I}^{*l})} \\ &= \alpha \frac{p(\mathbf{I}^{*l} \mid \mathbf{I}^l_M)}{p(\mathbf{I}^{*l})} p(\mathbf{I}^{*l} \mid \mathbf{I}^l_1, \ldots, \mathbf{I}^l_{M-1}) \end{aligned} \tag{1}$$

where $\alpha = (p(\mathbf{I}^l_1, \ldots, \mathbf{I}^l_{M-1})\, p(\mathbf{I}^l_M))/(p(\mathbf{I}^l_1, \ldots, \mathbf{I}^l_M))$ is a normalization constant term. In this regard, by considering $p(\mathbf{I}^{*l} \mid \mathbf{I}^l_1, \ldots, \mathbf{I}^l_M)$ as a Gaussian distribution with mean denoted as $\hat{\mu}^l_M$ and standard division denoted as $\hat{\sigma}^{l^2}_M$, we have

$$\begin{aligned} &\log p(\mathbf{I}^{*l} \mid \mathbf{I}^l_1, \ldots, \mathbf{I}^l_M) \\ &= \log\left(\alpha \frac{p(\mathbf{I}^{*l} \mid \mathbf{I}^l_M)}{p(\mathbf{I}^{*l})} p(\mathbf{I}^{*l} \mid \mathbf{I}^l_1, \ldots, \mathbf{I}^l_{M-1})\right) \\ &= \log p(\mathbf{I}^{*l} \mid \mathbf{I}^l_M) + \log p(\mathbf{I}^{*l} \mid \mathbf{I}^l_1, \ldots, \mathbf{I}^l_{M-1}) \\ &\quad - \log p(\mathbf{I}^{*l}) + \text{const} \end{aligned} \tag{2}$$

where "const", that is, $\log(\alpha)$, refers to the irrelevant term to $\mathbf{I}^{*l}$. According to [65], by assuming $p(\mathbf{I}^{*l} \mid \mathbf{I}^l_1)$ a non-informative prior, that is, $\mathcal{N}(\mu_0, \sigma^2_0 \longrightarrow \infty)$, the posterior distribution of $(\mathbf{I}^{*l} \mid \mathbf{I}^l_1, \ldots, \mathbf{I}^l_M)$ is a Gaussian distribution, $\mathcal{N}(\hat{\mu}^l_M, \hat{\sigma}^{l^2}_M)$, with lower uncertainty $(\hat{\sigma}^{l^2}_M < \sigma^2_0)$. Therefore, we can infer that the encoded samples, $\mathbf{I}'^l_1, \ldots, \mathbf{I}'^l_M$, belong to a Gaussian distribution that represents the optimal latent code, $\mathbf{I}^{*l}$, in the embedding space. By considering the assumed Gaussian distribution, we estimate the posterior distribution of $\mathcal{I}'^l$ via maximum likelihood estimation (MLE) [77]. In order to address that, we first convert $\mathcal{I}'^l$ to a set of "similarity" scores $(\mathcal{S}^l)$ to have more compact representation for the MLE. The similarity score for each sample, $m = 1, \ldots, M$, is obtained by (3)

$$s^l_m = \rho(\mathbf{I}'^l_m, \mathbf{I}''^l) \tag{3}$$

where s^l_m is the mth element in $\mathcal{S}^l$, and $\rho(\mathbf{I}'^l_m, \mathbf{I}''^l)$ is the Pearson correlation function for the given vectors, $\mathbf{I}'^l_m$ and $\mathbf{I}''^l$,

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defined in (4)

$$\rho(\mathbf{I}_m^{'l}, \mathbf{I}^{''l}) = \frac{\sum_{j=1}^{D'} \left(I_{jm}^{'l} - \bar{I}_m^{'l}\right)\left(I_j^{''l} - \bar{I}^{''l}\right)}{\left\{\sum_{j=1}^{D'} \left(I_{jm}^{'l} - \bar{I}_m^{'l}\right)^2 \sum_{j=1}^{D'} \left(I_j^{''l} - \bar{I}^{''l}\right)^2\right\}^{1/2}}. \quad (4)$$

For 3 and 4, $\mathbf{I}^{''l}$ is a mean of feature vectors obtained from 5

$$\mathbf{I}^{''l} = \left[\frac{1}{M}\sum_{m=1}^{M} I_{mj}^{'l}\right]_{j=1}^{D'} \quad (5)$$

where D' is the dimension of the encoded feature, $I_{mj}^{'l}$ is the jth, for $(j = 1, \ldots, D')$, feature value of the mth, for $(m = 1, \ldots, M)$, encoded sample. After generating $\mathcal{S}^l = \{s_m^l\}_{m=1}^M$, the probability distribution of the class l is determined by 6

$$\hat{p}^l\left(\mathcal{S}^l; \hat{\mu}_M^l, \hat{\sigma}_M^{l^2}\right) = \left[\frac{1}{\sqrt{2\pi}\hat{\sigma}^l} \exp\left(-\frac{1}{2\hat{\sigma}^{l^2}}(s_m^l - \hat{\mu}^l)^2\right)\right]_{m=1}^{M} \quad (6)$$

where $\hat{\mu}_M^l, \hat{\sigma}_M^{l^2}$ are the mean and variance estimated from the probability density function, denoted as $\hat{f}^l$, after receiving M samples. Here, $\hat{\mu}_M^l$ and $\hat{\sigma}_M^{l^2}$ are estimated using the MLE from 7

$$\{\hat{\mu}_M^l, \hat{\sigma}_M^{l^2}\} = \arg\max \prod_{m=1}^{M} \hat{f}^l\left(s_m^l \mid \{\hat{\mu}_M^l, \hat{\sigma}_M^{l^2}\}\right). \quad (7)$$

Finally, $\hat{p}^l$ receives the label of $\mathcal{I}^{'l}$ samples denoted[6] as y^l. The procedure of estimating the probability density function is summarized in Algorithm 2.

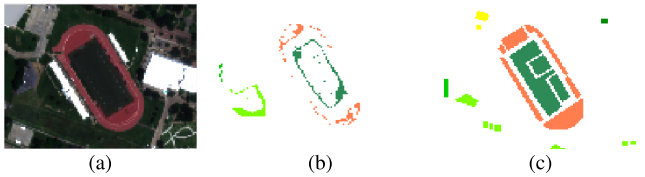
Algorithm 2 Estimate the Posterior Distribution of the Labeled Samples in Each Class.

Input: Given $\mathcal{I}^l = \{\mathbf{I}_m^l\}_{m=1}^M$; y^l.
Output: Estimate $\{\hat{p}^l\}_{l=1}^L$.

```
for l = 1, ..., L do
    for m = 1, ..., M do
        Obtain s_m^l using 3
        Generate S^l = {s_m^l}_{m=1}^M
    end for
    Estimate μ̂_M^l and σ̂_M^{l^2} using 7.
    Estimate p̂^l(S^l; μ̂_M^l, σ̂_M^{l^2}) using 6.
    Assign y^l to p̂^l(S^l; μ̂_M^l, σ̂_M^{l^2})
end for
```

2) Unlabeled Sample Matching: In this section, our aim is to measure the probability of $\mathcal{I}^{'u}$ samples matching the estimated distribution, $\hat{p}^l$. Accordingly, we first extract unlabeled sample patches whose centered samples belong to the label sets. The size of the patches extracted from the encoded unlabeled samples is denoted as $w' \times w' \times D'$. The extracted samples are then defined as $\hat{\mathcal{I}}^{'u} = \{\hat{\mathbf{I}}^{'u}\}_{n=1}^{N'}$, where $N' < N$, which contains encoded unlabeled samples in the spatial neighborhood of the labeled samples. Consequently, for the each of labeled classes l, the similarity scores of $\hat{\mathcal{I}}^{'u}$ samples

[6]Note that in the corresponding label set $(\mathcal{Y}^l)$, $y_m^l = y^l$ for $m = 1, \ldots, M$, and as a result, here we use y^l for the sake of simplicity.



(a) (b) (c)

Fig. 5. Example of pseudo-label generation of the Res-CP. (a) Cropped region from UH dataset. (b) Generated Res-CP pseudo-labels. (c) Ground truth $(\mathcal{G})$ of the dataset.

are obtained by using $\rho(\hat{\mathbf{I}}_n^{'u}, \mathbf{I}^{''l})$ for $n = 1, \ldots, N'$ with the similarity definition calculated by 4. The similarity scores represent the probability of each unlabeled sample matching the labeled samples' distribution. We then use the confidence interval range, denoted as R^l, to select samples with the highest matching probability. The selected samples receive the distribution label y^l. The steps of label generation for unlabeled samples are summarized in Algorithm 3. Specifically, for each unlabeled sample, $\hat{\mathbf{I}}_n^{'u}$, the similarity score s_n^u is obtained first. Then, if the value of s_n^u is greater than an $(R^l/2)$ confidence range, the nth sample of $\mathcal{I}^{'u}$ receives y^l. The labels and their corresponding image patches are stored in the pseudo-label set (defined as $\hat{\mathcal{Y}}^l$) and pseudo-image set (defined as $\hat{\mathcal{I}}^{'l}$), respectively.

Algorithm 3 Pseudo-Label Generation of the Encoded Unlabeled Samples

Input: Given $\hat{\mathcal{I}}^{'u} = \{\hat{\mathbf{I}}_n^{'u}\}_{n=1}^{N'}$; $\hat{p}^l$ with the label denoted as y^l, and confidence interval denoted as R^l.
Output: $\{\hat{\mathcal{Y}}^l\}_{l=1}^L$, $\{\hat{\mathcal{I}}^{'l}\}_{l=1}^L$.

```
for l = 1, ..., L do
    Set Ŷ^l = ∅
    Set Î^{'l} = ∅
    for n = 1, ..., N' do
        Obtain s_n^u = ρ(Î_n^{'u}, I^{''l}) using 3.
        if s_n^u > R^l/2 ⋀ s_n^u < R^l/2 then
            Add y^l to {Ŷ^l}
            Add Î_n^{'u} to {Î^{'l}}
        end if
    end for
end for
```

After generating $\hat{\mathcal{Y}}^l$ and $\hat{\mathcal{I}}^{'l}$, we, respectively, add $\mathcal{Y}^l$ and $\mathcal{I}^l$ to them in order to construct a larger training set. Fig. 5 provides an example of the pseudo-label generation results with respect to the original ground truth $(\mathcal{G})$.

C. Classification of HSIs Using PRCLSTM II

As mentioned earlier, the generated pseudo-samples can be used to enhance the learning of various DNNs to improve their classification results. In fact, the proposed framework is independent of any DNN architectures, because its main objective is to generate reliable pseudo-labels for unlabeled samples. Additionally, as a baseline DNN, we develop a classification model called PRCLSTM II in this article that effectively learns from pseudo-samples to classify the HSI samples. PRCLSTM II is a new version of the classification model

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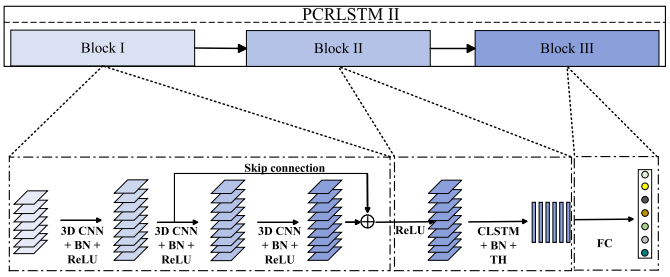


Fig. 6. PRCLSTM II framework.

(PRCLSTM) proposed in [13]. Originally, PRCLSTM was a patch-related convolution long short-term memory (CLSTM) network that consisted of three main blocks. Block I contained 3-D CNN layers, batch normalization (BN), and a rectified linear unit (ReLU) to perform spectral feature extraction. Block II consisted of a CLSTM layer, BN, and a hyperbolic tangent (HT) to recurrently perform spatial feature extraction. Finally, the extracted spectral–spatial features were fed to an FC layer in block III to perform probability estimation using softmax. More details regarding the learning configuration and model setup can be found in [13]. In this study, we further enhance the learning process of PRCLSTM by equipping its first block with residual learning. In particular, consider $\mathcal{H}^{B1}(\hat{\mathcal{I}}^{'l})$ to be the underlying mapping function of Block I, where $\hat{\mathcal{I}}^{'l}$ is the input of its layers. By applying the residual function such that $\mathcal{H}^{B1}(\hat{\mathcal{I}}^{'l}) - \hat{\mathcal{I}}^{'l}$, the underlying mapping function is updated to $\mathcal{H}^{B1}(\hat{\mathcal{I}}^{'l}) + \hat{\mathcal{I}}^{'l}$.

The key idea behind applying residual learning is to prevent gradient saturation and, as a result, enhance the learning capacity of the network [10]. The residual learning approach attempts to address the problem of performance degradation in classification models with several stacked nonlinear layers using the skip connection [60]. The skip connection provides a shortcut to propagate the gradient error to the layers of the network, and as a result, more effectively update their parameters [10].

The framework of PRCLSTM II is illustrated in Fig. 6. As can be seen from this figure, Block I is now equipped with residual learning with the addition of a 3-D CNN layer and a skip connection. The hyperparameters of PRCLSTM II are summarized in Table I.

IV. Experimental Results

In this section, the benchmark datasets, configurations, and assessment criteria utilized in our experiments are discussed. We then analyze the efficacy of the pseudo-samples in improving the accuracy results of the classification models. In addition, we provide a comprehensive comparison among the proposed PRCLSTM II classification model and its counterparts. Last, the Res-CP framework is analyzed by assessing the quantity and quality of the pseudo-samples, the influence of parameters, and the computational burden. All the experiments in this article are carried out using a graphics processing unit (GPU) on a laptop with 8 GB Geforce RTX 2080.

TABLE I
Configuration of the PRCLSTM II Hyperparameters

Layer Name/ Input shape	Kernel #/ Size	Components
$\begin{bmatrix} \text{Block I-1} \\ w \times w \times C \end{bmatrix}$	$\begin{bmatrix} \text{CNN, } 24 \\ 1 \times 1 \times 7 \end{bmatrix} \times 3$	BN, ReLU, Add
$\begin{bmatrix} \text{Block I-2} \\ w \times w \times C^1 \times 24 \end{bmatrix}$	$\begin{bmatrix} \text{CNN, } 128 \\ 1 \times 1 \times C^1 \end{bmatrix}$	BN, ReLU
$\begin{bmatrix} \text{Block II} \\ w \times w \times 1 \times 128 \end{bmatrix}$	$\begin{bmatrix} \text{CLSTM, } 18 \\ w \times w \times 1 \times 128 \end{bmatrix}$	BN, HT
$\begin{bmatrix} \text{Block III} \\ 18 \end{bmatrix}$	[FC, L]	Softmax

A. Dataset Description

Four commonly used datasets having various spectral and spatial characteristics are utilized in our experiments. For each dataset, we randomly select ten samples per class as a training set and reserve the rest of the labeled samples as a test set for the assessment and evaluation tasks. The description of each dataset is summarized below:

1) **PaviaU**: Pavia University (PaviaU) is a publicly available dataset, collected using the Reflective Optics System Imaging Spectrometer (ROSIS-3) in 2001. The spatial dimension and spatial resolution of PaviaU are equal to 610 × 340 and 1.3 m, respectively. After the noise and outliers removal procedure, the remaining 103 spectral bands in the range of 0.43–0.86 μm are applied in our experiments. The corresponding ground truth (GT) contains 42 776 labeled pixels categorized into nine classes from land covers of the scene. Table II shows the number of samples in the training and test sets.
2) **UH**: University of Houston (UH) [7] [78] is a publicly available large-scale dataset, originally utilized in the IEEE data fusion contest in the year 2013. The dataset was collected by the National Center for Airborne Laser Mapping (NCALM) and has a spatial size of 349 × 1905 and a spatial resolution of 2.5 m. The UH dataset has 144 spectral bands over the 0.36–1.05 μm wavelength range. As shown in Table II, 15 various land cover classes are available in the ground truth of the dataset.
3) **SA**: The third publicly available dataset used in this study is Salinas (SA), which was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) in 1992. The spatial dimension of the SA is 512 × 217 pixels and the spatial resolution is 3.7 m. After noise removal, the remaining 204 spectral bands covering the wavelength range of 0.4–2.5 μm are used for our experiments. The corresponding GT contains 54 129 samples in 16 land cover classes. Table II tabulates the number of samples randomly selected for training and testing tasks.
4) **KSC**: The last dataset applied in our experiments is Kennedy Space Center (KSC), collected by AVIRIS in 1996. This publicly available dataset contains

[7] www.hyperspectral.ee.uh.edu

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TABLE II

LAND COVER CLASS INFORMATION AND THE NUMBER OF ANNOTATED SAMPLES IN THE UNIVERSITY OF PAVIA (PAVIAU), UH, SA, AND KSC DATASETS

	PaviaU			UH			SA			KSC		
No.	Class	Train	Test	Class	Train	Test	Class	Train	Test	Class	Train	Test
1	Asphalt	10	6621	Healthy grass	10	1241	Brocoli-green-weeds-1	10	1999	Scrub	10	751
2	Meadows	10	18639	Stressed grass	10	1244	Brocoli-green-weeds-2	10	3716	willow swamp	10	233
3	Gravel	10	2089	Synthetic grass	10	687	Fallow	10	1966	CP hammock	10	246
4	Trees	10	3054	Trees	10	1234	Fallow-rough-plow	10	1384	Slash Pine	10	242
5	Painted metal sheets	10	1335	Soil	10	1232	Fallow-smooth	10	2668	Oak/Board leaf	10	151
6	Bare Soil	10	5019	Water	10	315	Stubble	10	3949	Hardwood	10	219
7	Bitumen	10	1320	Residential	10	1258	Celery	10	3569	Swap	10	95
8	Self-Blocking Bricks	10	3672	Commercial	10	1234	Grapes-untrained	10	11261	Graminoid marsh	10	421
9	Shadows	10	937	Road	10	1242	Soil-vinyard-develop	10	6193	Spartina marsh	10	510
10	-	-	-	Highway	10	1217	Corn-senesced-green-weeds	10	3268	Cattail marsh	10	394
11	-	-	-	Railway	10	1225	Lettuce-romaine-4wk	10	1058	Salt marsh	10	409
12	-	-	-	Parking Lot 1	10	1223	Lettuce-romaine-5wk	10	1917	Mud flats	10	493
13	-	-	-	Parking Lot 2	10	459	Lettuce-romaine-6wk	10	906	Water	10	917
14	-	-	-	Tennis Court	10	418	Lettuce-romaine-7wk	10	1060	-	-	-
15	-	-	-	Running Track	10	650	Vinyard-untrained	10	7258	-	-	-
16	-	-	-	-	-	-	Vinyard-vertical-trellis	10	1797	-	-	-
Total		**90**	**42686**		**150**	**14879**		**160**	**44485**		**130**	**5081**
Total by %		**0.2%**	**99.8%**		**1%**	**99%**		**0.4%**	**99.6%**		**2.5%**	**97.5%**

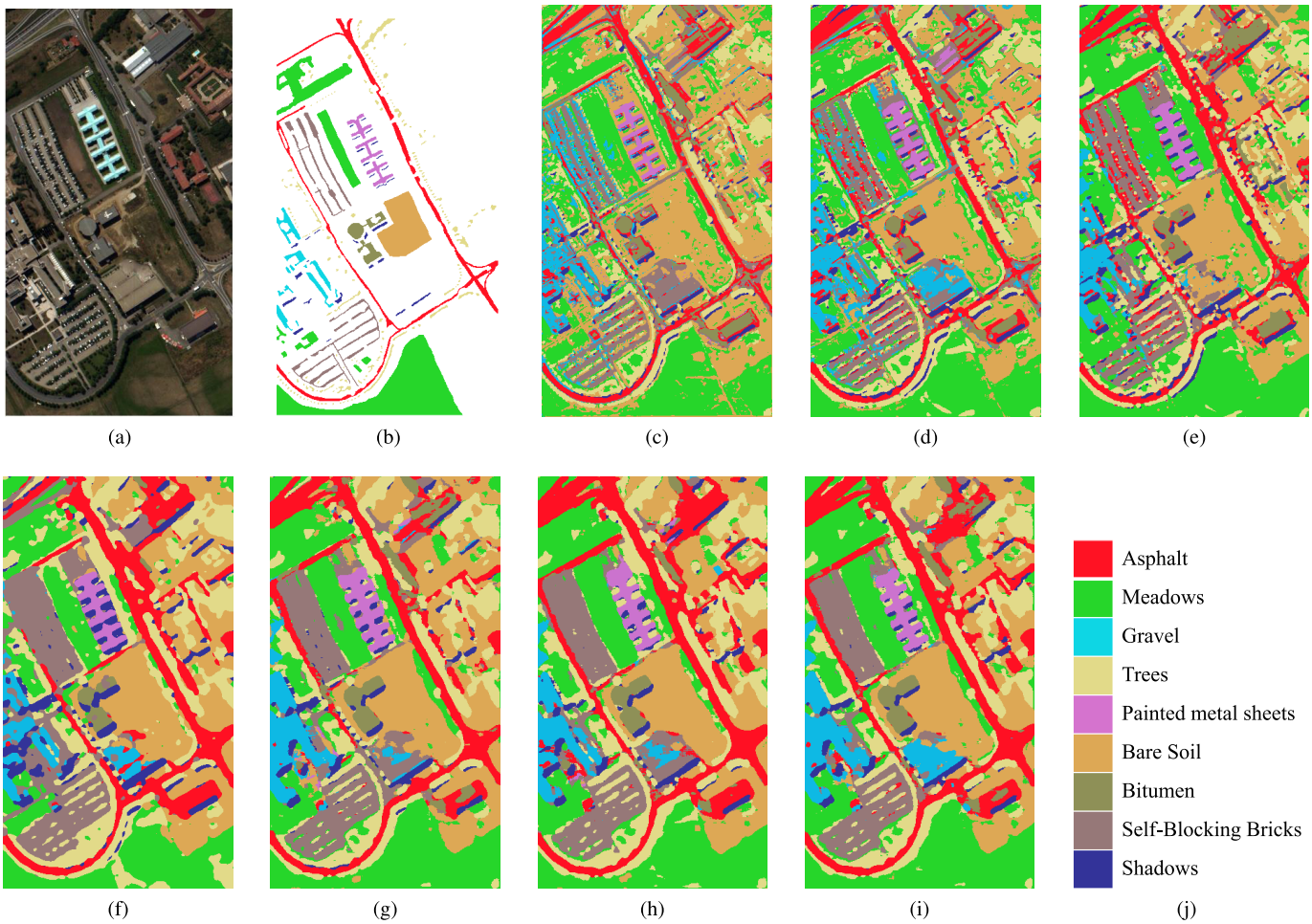


Fig. 7. Classification maps of the Pavia University (PaviaU) dataset. (a) PaviaU false color composite. (b) PaviaU ground truth. (c) 3-D CNN. (d) CRNN. (e) Fast-3-D. (f) SS-SEN. (g) PRCLSTM. (h) SSRN. (i) PRCLSTM II. (j) Figure legend.

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TABLE III
NUMBER OF PSEUDO-SAMPLES GENERATED BY PLG

UP	UH	SA	KSC
5710 ± 144	7683 ± 316	5056 ± 164	1328 ± 90

TABLE IV
CONFIGURATION OF THE RCN ENCODING NETWORK

Layer Name/ Input shape	Kernel #/ Size	Components
$\begin{bmatrix} \text{Res-Conv-1} \\ w \times w \times C \end{bmatrix}$	$\begin{bmatrix} \text{CNN, } 24 \\ 1 \times 1 \times 7 \end{bmatrix} \times 3$	BN, P-ReLu, Add
$\begin{bmatrix} \text{Conv-1} \\ w \times w \times C^1 \times 24 \end{bmatrix}$	$\begin{bmatrix} \text{CNN, } 128 \\ 1 \times 1 \times C^1 \end{bmatrix}$	BN, P-ReLu
$\begin{bmatrix} \text{Reshape} \\ w \times w \times 1 \times 128 \end{bmatrix}$	-	-
$\begin{bmatrix} \text{Res-Conv-2} \\ w \times w \times 128 \times 1 \end{bmatrix}$	$\begin{bmatrix} \text{CNN, } 24 \\ 3 \times 3 \times 128 \end{bmatrix} \times 3$	BN, P-ReLu, Add
$\begin{bmatrix} \text{Average pooling} \\ w^1 \times w^1 \times 1 \times 24 \end{bmatrix}$	$w^1 \times w^1$	-
$\begin{bmatrix} \text{Center coordinates} \\ 1 \times 1 \times 1 \times 24 \end{bmatrix}$	FC, 2	BN, P-ReLu
$\begin{bmatrix} \text{Class probability} \\ 2 \end{bmatrix}$	FC, L	Softmax

614×512 pixels with a spatial resolution of 18 m. The original spectral dimension of the KSC shrank from 224 to 176 over the range of 0.4–2.5 μm after bands affected by noise and water absorption were removed. The dataset has 5211 annotated labels in 13 land cover classes. Table II tabulates the training and testing sets randomly selected for our experiments.

B. Experimental Configurations and Assessment Criteria

As mentioned earlier, the Res-CP framework consists of two main subcomponents, RCN and PLG, to generate the pseudo-samples.

1) RCN Configuration: As shown in Table IV, the number of CNN kernels is set inspired by [60]. The learning rate is fixed to 0.0003, also based on [60]. The batch size and number of epochs are set to 16 and 500, respectively. The hyperparameters of the center-softmax loss, λ and α, which control the intraclass variation and learning rate of centers, are fixed to 0.1 and 0.5, respectively. The root mean squared error propagation (RMSProp) [79] is used to optimize the center-softmax loss functions. For the training task, w is fixed to 9 and the softmax layer is tuned according to the number of available classes in each of the datasets. As previously mentioned, ten samples per class are randomly selected from the datasets in question, so that the total number of training sets is 90 (9×10), 150 (15×10), 160 (16×10), and 130 (13×10) for PaviaU, UH, SA, and KSC, respectively. The remaining samples are reserved for the testing task. It is worth mentioning that no validation set is selected from the original GT of the datasets for any experiments conducted in this article.

2) PLG Configuration: The configurations of two hyperparameters associated with the PLG setup, namely R^l and w', are described in this section. R^l, which represents the confidence interval for the matching task, is fixed at 0.5. w' is set by considering each dataset's spatial resolution. In other words, w' is set in such a way that it extracts unlabeled samples with roughly the same spatial distance to the labeled samples in all datasets in question. To this end, the w' values are, respectively, set to 20, 10, 10, and 5 for PaviaU, UH, SA, and KSC datasets.

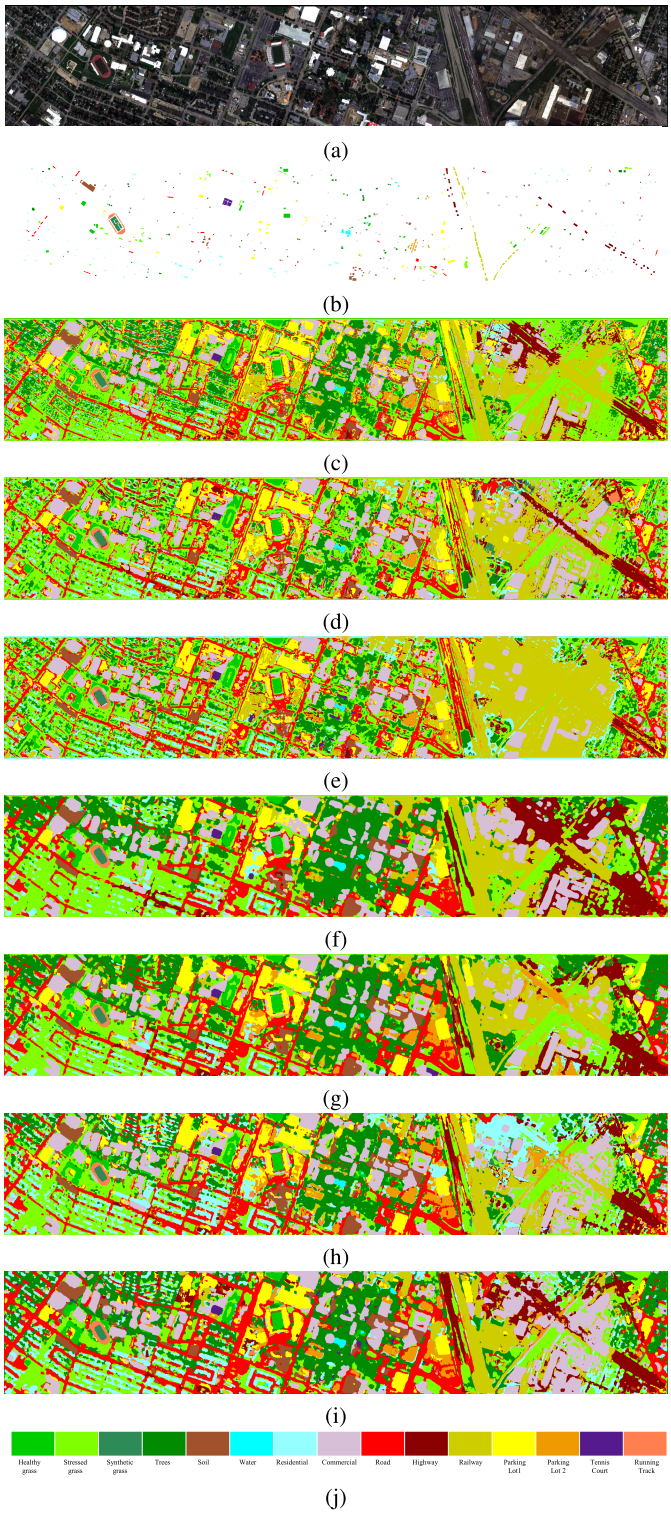


Fig. 8. Classification maps of the UH dataset. (a) UH false color composite. (b) UH ground truth. (c) 3-D CNN. (d) CRNN. (e) Fast-3-D. (f) SS-SEN. (g) PRCLSTM. (h) SSRN. (i) PRCLSTM II. (j) Figure legend.

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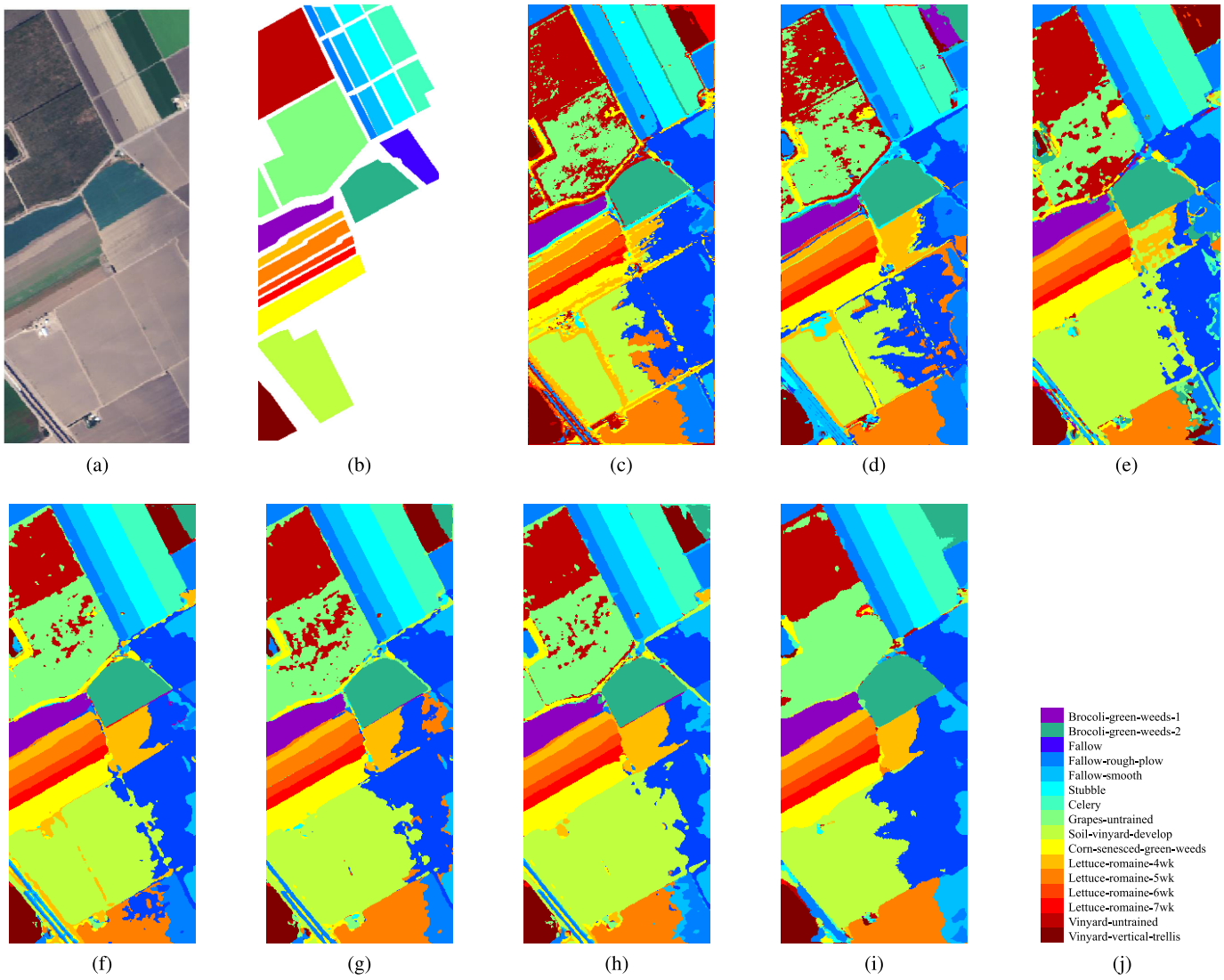


Fig. 9. Classification maps of the SA dataset. (a) SA false color composite. (b) SA ground truth. (c) 3-D CNN. (d) CRNN. (e) Fast-3-D. (f) SS-SEN. (g) PRCLSTM. (h) SSRN. (i) PRCLSTM II. (j) Figure legend.

3) Assessment Criteria: Three well-known quantitative measurements, namely overall accuracy (OA), kappa coefficient (κ), and average accuracy (AA), are taken into account. For each experiment, five runs are executed to obtain the mean and standard deviation of the results.

C. Evaluation of the Pseudo-Samples of Res-CP

In order to quantitatively and qualitatively assess the pseudo-samples generated by the Res-CP framework, two sets of experiments are carried out. In the first set, the aim is to analyze the impact of Res-CP pseudo-samples in improving the generalization capacity of the classification model. To do that, the classification models are trained in two scenarios—**Scenario (a)**: Train from scratch using initial training samples, that is, $\mathcal{I}^l$ and $\mathcal{Y}^l$, described in Table II; and **Scenario (b)**: Train from scratch using the Res-CP pseudo-samples, that is, $\hat{\mathcal{I}}'^l$ and $\hat{\mathcal{Y}}^l$. The number of generated pseudo-samples for the experimental datasets is tabulated in Table III.

Regarding the classification models, six state-of-the-art DNNs having various architectures such as CNN, RNN, and residual connections are applied. The details of these six models are described as follows:

1) 3-D CNN [80]: A 3-D spectral–spatial classification model that contains several 3-D CNN layers followed by FC layers is used.
2) Fast-3-D [81]: A fast 3-D CNN model is used to perform a classification task with a light computational burden.
3) Convolution Recurrent Neural Network (CRNN) [39]: A hybrid CNN and RNN model, equipped with 1-D CNN and a long short-term memory (LSTM) unit, is used to extract spectral features from the pseudo-labels. The model also includes a spatial constraint technique to take spatial information into consideration.
4) Spectral-Spatial Squeeze-and-Excitation Network (SS-SEN) [74]: A 3-D spectral–spatial residual CNN model enhanced by bag-of-features and squeeze-and-excitation techniques is utilized to perform the classification task.
5) PRCLSTM [13]: A 3-D CNN-RNN model is considered, which includes a CLSTM unit to extract relational

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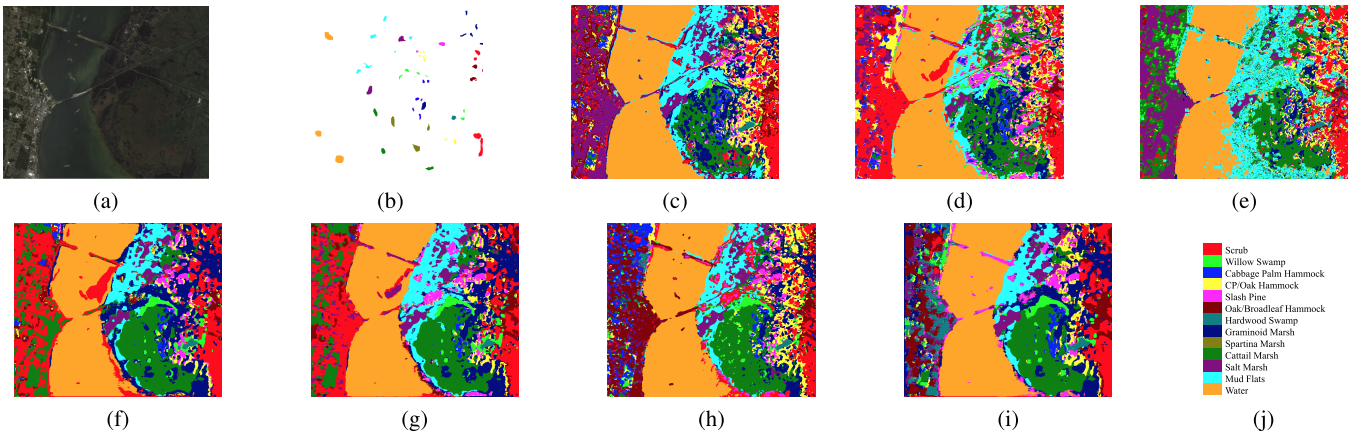


Fig. 10. Classification maps of the KSC dataset. (a) KSC false color composite. (b) KSC ground truth. (c) 3-D CNN. (d) CRNN. (e) Fast-3-D. (f) SS-SEN. (g) PRCLSTM. (h) SSRN. (i) PRCLSTM II. (j) Figure legend.

TABLE V

CLASSIFICATION ACCURACY (%) RESULTS OF THE HSI DATASETS IN QUESTION. (A) **SCENARIO (A)**. (B) **SCENARIO (B)**

	(A)					(B)				
Model	Metric	PaviaU	UH	SA	KSC	Metric	PaviaU	UH	SA	KSC
3D-CNN	OA	63.6 ± 0.59	50.65 ± 0.17	71.16 ± 1.59	82.1 ± 0.14	OA	**80.22 ± 1.96**	**77.19 ± 0.94**	**87.71 ± 0.4**	**93.82 ± 0.15**
	κ	53.13 ± 0.63	46.63 ± 0.18	68.21 ± 1.7	80.07 ± 0.16	κ	**84.59 ± 1.66**	**75.33 ± 1.02**	**86.4 ± 0.42**	**93.12 ± 0.16**
	AA	63.8 ± 1.32	57.02 ± 0.1	73.39 ± 1.14	75.92 ± 0.3	AA	**82.2 ± 1.69**	**82.39 ± 0.51**	**90.16 ± 0.42**	**91.89 ± 0.53**
CRNN	OA	64.11 ± 0.48	68.13 ± 0.55	80.11 ± 2.8	76.27 ± 0.42	OA	**89.55 ± 1.02**	**77.39 ± 1.4**	**88.82 ± 0.47**	**91.69 ± 0.39**
	κ	55.6 ± 0.57	65.64 ± 0.61	78.01 ± 3	73.58 ± 0.47	κ	**86.38 ± 1.27**	**75.57 ± 1.52**	**87.62 ± 0.53**	**90.74 ± 0.44**
	AA	66.96 ± 0.27	68.16 ± 1.25	86.02 ± 1.18	69.72 ± 0.26	AA	**85.89 ± 1.29**	**80.63 ± 1.03**	**90.89 ± 0.81**	**89.87 ± 0.37**
Fast-3D	OA	42.44 ± 12.05	56.82 ± 0.77	77.44 ± 2.07	76.14 ± 0.63	OA	**92.96 ± 1.47**	**72.28 ± 2.75**	**91.03 ± 1.49**	**81.36 ± 0.78**
	κ	29.78 ± 10.32	53.65 ± 0.85	75.09 ± 2.26	73.48 ± 0.67	κ	**90.74 ± 1.9**	**70.02 ± 3**	**90.07 ± 1.62**	**79.27 ± 0.88**
	AA	20.23 ± 4.56	58.84 ± 0.79	80.57 ± 2.79	69.58 ± 0.96	AA	**90.09 ± 2.9**	**80.91 ± 1.06**	**93.11 ± 1.09**	**79.34 ± 1.2**
CB-SS	OA	61.33 ± 10.83	31.73 ± 1.97	77.55 ± 15.97	49.12 ± 10.69	OA	**94.98 ± 0.96**	**80.52 ± 2.74**	**93.59 ± 0.66**	**91.85 ± 3.39**
	κ	54.47 ± 10.67	25.89 ± 2.19	75.58 ± 16.77	44.37 ± 11.23	κ	**93.43 ± 1.23**	**78.92 ± 2.98**	**92.9 ± 0.73**	**90.91 ± 3.8**
	AA	64.95 ± 7.2	26.6 ± 3.52	78.34 ± 11.53	44.61 ± 10.52	AA	**93.6 ± 0.85**	**79.55 ± 6.02**	**96.56 ± 0.56**	**91.76 ± 2.47**
PRCLSTM	OA	79.07 ± 2.12	54 ± 2.02	84.24 ± 0.68	90.8 ± 0.92	OA	**95.17 ± 1.55**	**79.89 ± 1.13**	**94.09 ± 2.00**	**93.98 ± 1.21**
	κ	72.75 ± 2.45	50.59 ± 2.11	82.54 ± 0.7	89.75 ± 1.03	κ	**93.68 ± 2**	**78.29 ± 1.21**	**93.44 ± 2.00**	**93.29 ± 1.35**
	AA	76.27 ± 1.71	61.64 ± 1.82	90.53 ± 0.84	86.63 ± 1.02	AA	**93.13 ± 1.76**	**83.24 ± 1.11**	**96.79 ± 0.42**	**90.99 ± 0.77**
SSRN	OA	83.1 ± 0.74	79.54 ± 0.58	88.1 ± 1.68	94.61 ± 0.35	OA	**95.58 ± 0.99**	**83.36 ± 0.79**	**95.6 ± 1.13**	**96.92 ± 0.3**
	κ	78.72 ± 0.86	77.9 ± 0.62	86.86 ± 1.83	93.99 ± 0.39	κ	**94.2 ± 1.28**	**82.02 ± 0.85**	**95.1 ± 1.25**	**96.57 ± 0.33**
	AA	82.21 ± 0.54	82.32 ± 0.74	94.83 ± 0.26	93.86 ± 0.37	AA	**93.99 ± 1.2**	**85.69 ± 1.31**	**97.33 ± 0.35**	**95.13 ± 0.33**
PRCLSTM II	OA	77.22 ± 1.61	75.48 ± 1.39	86.21 ± 0.93	90.55 ± 0.74	OA	**96.62 ± 1.11**	**85.57 ± 1.45**	**97.24 ± 0.54**	**96.94 ± 0.34**
	κ	70.78 ± 1.85	73.51 ± 1.51	84.65 ± 1.02	89.48 ± 0.82	κ	**95.56 ± 1.44**	**84.43 ± 1.56**	**96.93 ± 0.6**	**96.59 ± 0.37**
	AA	75.52 ± 1.2	78.42 ± 1.17	91.19 ± 0.26	86.63 ± 0.55	AA	**95.03 ± 1.54**	**87.02 ± 1.13**	**97.46 ± 0.86**	**95.09 ± 0.44**

spectral and spatial information from the sequence of HSI patches.

6) SSRN [60]: A 3-D residual CNN model enhanced by two residual blocks is employed to separately extract spectral and spatial features for HSI classification.

For the training tasks, the hyperparameters of the classification models are set according to their original settings and trained for 200 epochs. Since most of the classification models are equipped with regularization components, 30% of the pseudo-samples are considered to be the validation samples to perform the regularization tasks.

After performing the training tasks, the training models in Scenarios (a) and (b) are assessed based on the test sample sets. The test accuracy assessment results for Scenarios (a) and (b) are tabulated in sections (a) and (b) of Table V, respectively. The results demonstrate that the Res-CP pseudo-samples can significantly improve the accuracy results of classification models up to 50.52% OA, 60.96% κ, and 69.86% AA. According to the results, training with the Res-CP pseudo-samples yields significant improvements in shallow models and also enables deeper classification models to achieve accurate results compatible with state-of-the-art supervised learning results. As an example of a shallow DNN model, Fast-3-D yields a very significant improvement in classification performance from 42.44% OA in Scenario (a) to 92.96% in Scenario (b) for the PaviaU dataset. The improvement can also be seen in the deeper models such as

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TABLE VI
CLASSIFICATION ACCURACY (%) OF THE COMPETING CLASSIFICATION MODELS FOR THE UNIVERSITY OF PAVIA (PAVIAU) DATASET WITH RES-CP PSEUDO-SAMPLES

No.	3D-CNN	CRNN	Fast-3D	SS-SEN	PRCLSTM	SSRN	PRCLSTM II
1	96.73 ± 0.36	95.9 ± 0.88	98.62 ± 0.67	99.47 ± 0.23	99.33 ± 0.21	99.46 ± 0.11	99.18± 0.45
2	97.75 ± 0.52	97.76 ± 0.28	97.75 ± 0.54	99.57 ± 0.16	99.47 ± 0.44	99.76 ± 0.05	99.43± 0.05
3	63.53 ± 2.41	80.35 ± 1.94	85.24 ± 8.74	93.63 ± 1.81	91.74 ± 7.64	96.74 ± 0.71	98.7 ±1.4
4	90.31 ± 10.53	79.42 ± 8.81	80.14 ± 5.78	79.02 ± 6.28	83.63 ± 13.27	80.24 ± 6.05	91.1± 3.46
5	99.19 ± 0.57	99.56 ± 0.31	98.55 ±1.23	96.66 ± 0.94	96.44 ± 0.78	98.33 ± 0.73	95.53± 1.83
6	65.25 ± 3.13	85.87 ± 1.52	93.17 ± 3.23	92.33 ± 3.72	93.18 ± 5.74	98.51 ± 0.79	94.22± 4.98
7	50.15 ± 4.04	57.92 ± 3.61	89.5 ± 4.61	98.75 ± 2.13	87.95 ± 5.85	91.3 ± 5.58	87.85± 5.78
8	76.87 ± 2.15	78.22 ± 1.52	82.68 ± 5.43	86.17 ± 3.32	89.7 ± 3.26	82.72 ± 3.84	90.48 ±3.27
9	99.97 ± 0.05	97.99 ± 0.18	85.17 ± 13.7	96.81 ± 2.11	96.74 ± 1.32	98.85 ± 0.43	98.73 ±0.68
OA	80.22 ± 1.96	89.55 ± 1.02	92.96 ± 1.47	94.98 ± 0.96	95.17 ± 1.55	95.58 ± 0.99	**96.62 ± 1.11**
κ	84.59 ± 1.66	86.38 ± 1.27	90.74 ± 1.9	93.43 ± 1.23	93.68 ± 2	94.2 ± 1.28	**95.56 ± 1.44**
AA	82.2 ± 1.69	85.89 ± 1.29	90.09 ± 2.9	93.6 ± 0.85	93.13 ± 1.76	93.99 ± 1.2	**95.03 ± 1.54**

TABLE VII
CLASSIFICATION ACCURACY (%) OF THE COMPETING CLASSIFICATION MODELS FOR THE UH DATASET WITH RES-CP PSEUDO-SAMPLES

No.	3D-CNN	CRNN	Fast-3D	SS-SEN	PRCLSTM	SSRN	PRCLSTM II
1	99.66 ± 0.45	99.1 ± 0.74	95.01 ± 8.56	95.36 ± 1.92	98.36 ± 0.97	99.28 ± 0.63	96.42 ± 1
2	67.54 ± 0.85	56.24 ± 2.22	77.61 ± 6.58	71.8 ± 4.17	67.4 ± 3.16	67.62 ± 0.64	75.63 ± 4.21
3	98.25 ± 0.7	95.66 ± 1.61	99.52 ± 0.62	88.13 ± 19.94	98.75 ± 1.11	93.61 ± 8.38	99.88 ± 0.23
4	81.74 ± 3.45	76.47 ± 3.93	72.4 ± 2.3	69.59 ± 11.38	71.58 ± 2.65	84.55 ± 7.34	90 ± 3.59
5	96.29 ± 0.83	97.25 ± 2.34	98.34 ± 1.01	95.05 ± 1.52	89.89 ± 5.37	99.68 ± 0.37	97.27 ± 2.79
6	94.11 ± 2.14	98.72 ± 0.37	92.87 ± 1.8	76.22 ± 38.59	98.94 ± 0.98	95.12 ± 6.68	97.98 ± 0.75
7	86.11 ± 1.16	89.1 ± 1.2	90.78 ± 1.33	82.17 ± 6.91	89.75 ± 2	83.5 ± 10.22	90.09 ± 5.65
8	76.8 ± 6.23	75.3 ± 7.31	93.28 ± 3.2	97.37 ± 2.36	91.05 ± 10.27	83.95 ± 8.09	76.01 ± 9.53
9	79.71 ± 3.11	84.5 ± 2.1	74.78 ± 2.6	70.3 ± 7.94	77.1 ± 1.54	81.62 ± 1.26	85.68 ± 4.49
10	63.37 ± 5.88	73.07 ± 3.99	55.14 ± 3.57	71.37 ± 2.08	74.09 ± 2.39	82.38 ± 3.45	86.41 ± 3.51
11	60.8 ± 4.22	74.86 ± 2.73	38.26 ± 2.54	82.86 ± 5.05	75.35 ± 4.13	80.73 ± 7.09	85.65 ± 5.38
12	52.15 ± 4.06	61.97 ± 3.79	55.53 ± 6.66	77.52 ± 5.59	70.96 ± 2.58	72.17 ± 2.25	78.45 ± 3.9
13	82.5 ± 6.2	67.68 ± 17.01	88.82 ± 2.63	92.15 ± 8.84	50.95 ± 6.59	64.9 ± 6.98	55.98 ± 10.69
14	99.9 ± 0.12	83.97 ± 3.87	81.31 ± 2.59	38.49 ± 47.16	98 ± 2.85	100 ± 0	95.04 ± 2.84
15	96.83 ± 0.97	75.44 ± 2.69	100 ± 0	84.88 ± 17.77	96.43 ± 0.53	96.3 ± 1.06	94.75 ± 2.18
OA	77.19 ± 0.94	77.39 ± 1.4	72.28 ± 2.75	80.52 ± 2.74	79.89 ± 1.13	83.36 ± 0.79	**85.57 ± 1.45**
κ	75.33 ± 1.02	75.57 ± 1.52	70.02 ± 3	78.92 ± 2.98	78.29 ± 1.21	82.02 ± 0.85	**84.43 ± 1.56**
AA	82.39 ± 0.51	80.63 ± 1.03	80.91 ± 1.06	79.55 ± 6.02	83.24 ± 1.11	85.69 ± 1.31	**87.02 ± 1.13**

SS-SEN and SSRN, which obtain 54.9% and 86.3% average OA in Scenario (a) in comparison with 90.2% and 92.9% in Scenario (b), for all the experimental datasets.

In addition to the quantitative analysis, the classical classification maps of trained models in scenario (b), that is, those trained with the Res-CP pseudo-samples, are demonstrated in Figs. 7–10. By comparing the classification maps with the ground truths, it can be observed that most of the competing methods, and in particular, deeper models such as PRCLSTM, SSRN, and PRCLSTM II, obtain reasonable classification maps.

In the second set of experiments, a more comprehensive comparison between the accuracy results of the proposed PRCLSTM II and its rivals in each class is performed. Tables VI–IX report the results of the competing methods on the test sets of PaviaU, UH, SA, and KSC datasets, respectively. It should be noted that all the methods are trained by using the generated pseudo-sample sets, that is, $\hat{\mathcal{I}}^l$ and $\hat{\mathcal{Y}}^l$. According to the results, the proposed PRCLSTM II demonstrated robust performance and outperformed its rivals based on most of the assessment criteria. Moreover, as can be seen from the classification maps in Figs. 7–10, the proposed PRCLSTM II can predict challenging and highly correlated land covers, such as the "Grapes-untrained" class in SA, and "Railway" and "Highway" classes in UH, more accurately. The proposed PRCLSTM II obtains 3.3%, 3.7%,

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TABLE VIII
CLASSIFICATION ACCURACY (%) OF THE COMPETING CLASSIFICATION MODELS FOR THE SA DATASET WITH RES-CP PSEUDO-SAMPLES

No.	3D-CNN	CRNN	Fast-3D	SS-SEN	PRCLSTM	SSRN	PRCLSTM II
1	98.12 ± 2.24	99.87 ± 0.12	99.86 ± 0.12	99.03 ± 1.41	99.22 ± 0.79	100 ± 0	99.64 ± 0.64
2	98.88 ± 0.26	99.66 ± 0.18	98.73 ± 1.08	99.96 ± 0.05	99.70 ± 0.88	99.73 ± 0.21	99.99 ± 0.02
3	89.62 ± 0.78	71.08 ± 2.3	90.23 ± 5.51	99.76 ± 0.43	96.22 ± 2.51	97.76 ± 0.94	98.44 ± 0.57
4	98.21 ± 1.38	90.13 ± 7.19	93.75 ± 5.05	94.44 ± 5.44	98.99 ± 0.89	98.27 ± 0.41	98.82 ± 0.95
5	98.87 ± 0.36	87.96 ± 4.32	94.96± 2.26	98.27 ± 1.03	98.71 ± 0.59	98.84 ± 1.6	96.01 ± 2.79
6	98.31 ± 0.61	97.65 ± 0.96	98.93 ± 1.04	99.85 ± 0.24	99.8 ± 0.22	99.96 ± 0.05	99.76 ± 0.13
7	99.82 ± 0.28	98.88 ± 0.46	96.31 ± 1.11	98.08 ± 2.3	99.55 ± 0.57	99.91 ± 0.18	99.72 ± 0.46
8	92.64 ± 4.89	89.54 ± 2.46	93.96 ± 1.52	97.02 ± 1.06	96.49 ± 2.51	93.07 ± 0.67	97.35 ± 1.45
9	96.66 ± 0.48	99.31 ± 0.55	98.47 ± 0.85	99.69 ± 0.15	99.91 ± 0.08	99.66 ± 0.13	99.65 ± 0.24
10	78.24 ± 0.73	82.44 ± 1.29	90.7 ± 4.61	95.48 ± 1.87	93.14 ± 1.18	94.43 ± 0.69	94.88 ± 2.12
11	48.62 ± 1.05	77.11 ± 4.28	90.2 ± 2.78	97.78 ± 0.87	96.65 ± 1.00	93.98 ± 0.83	87.33 ± 9.99
12	96.9 ± 0.47	98.33 ± 0.32	91.6 ± 6.01	99.69 ± 0.57	99.76 ± 0.61	97.86 ± 0.48	96.44 ± 4.75
13	87.87 ± 2.56	99.62 ± 0.18	96.01 ± 2.73	95.12 ± 5.13	95.50 ± 2.41	99.87 ± 0.26	99.44 ± 1.12
14	94.89 ± 1.66	99.2 ± 0.47	88.26 ± 10.94	98.88 ± 0.64	98.82 ± 1.22	98.57 ± 0.47	99.02 ± 1.17
15	65.38 ± 4.91	69.73 ± 1.16	72.4 ± 5.27	72.25 ± 3.21	76.13 ± 8.65	85.31 ± 6.92	92.88 ± 3.85
16	99.55 ± 0.15	93.7 ± 3.13	95.34 ± 4.53	99.69 ± 0.57	100 ± 0	100 ± 0	100 ± 0
OA	87.71 ± 0.4	88.82 ± 0.47	91.03 ± 1.49	93.59 ± 0.66	94.09 ± 2.00	95.6 ± 1.13	**97.24 ± 0.54**
κ	86.4 ± 0.42	87.62 ± .053	90.07 ± 1.62	92.9 ± 0.73	93.44 ± 2.00	95.1 ± 1.25	**96.93 ± 0.6**
AA	90.16 ± 0.42	90.89 ± 0.81	93.11 ± 1.09	96.56 ± 0.56	96.79 ± 0.42	97.33 ± 0.35	**97.46 ± 0.86**

TABLE IX
CLASSIFICATION ACCURACY (%) OF THE COMPETING CLASSIFICATION MODELS FOR THE KSC DATASET WITH RES-CP PSEUDO-SAMPLES

No.	3D-CNN	CRNN	Fast-3D	SS-SEN	PRCLSTM	SSRN	PRCLSTM II
1	98.95 ± 0.65	95.58 ± 0.48	95.09 ± 4.25	96.48 ± 3.86	99.49 ± 0.4	99.31 ± 0.78	99.92 ± 0.11
2	95.25 ± 1.64	81.09 ± 0.73	64.59 ± 10.61	99.32 ± 1.14	93.43 ± 2.68	96.39 ± 1.28	96.45 ± 2.99
3	99.59 ± 0.63	99.74 ± 0.21	87.48 ± 2.39	98.47 ± 2.84	100 ± 0	97.35 ± 0.93	92.43 ± 8.92
4	82.38 ± 4.18	88.59 ± 1.11	68.48 ± 6.22	86.49 ± 6.3	74.22 ± 3.2	80.4 ± 4.78	95.66 ± 5.03
5	85.7 ± 8.61	78.63 ± 2.93	70.47 ± 3.03	86.27 ± 4.61	49.75 ± 9.69	86.38 ± 3.4	78.72 ± 5.84
6	91.06 ± 2.25	97.89 ± 0.22	70.66 ± 3.3	89.63 ± 7.25	96.16 ± 3.42	95.65 ± 1.09	94.96 ± 2.59
7	81.68 ± 3.63	79.19 ± 1.89	75.73 ± 6.7	85.1 ± 5.77	81.12 ± 6.77	91.14 ± 2.99	89.06 ± 6.47
8	84.08 ± 2.7	80.24 ± 0.7	71.35 ± 3.02	69.46 ± 8.08	95.12 ± 3.92	94.42 ± 0.49	94.65 ± 3.62
9	98.56 ± 0.13	99.43 ± 0.18	99.55 ± 0.34	100 ± 0	94.52 ± 2.27	99.42 ± 1.16	97.7 ± 1.54
10	88.77 ± 3.38	94.34 ± 1.16	77.67 ± 4.36	94.87 ± 5.53	100 ± 0	98.55 ± 0.8	99.35 ± 1.31
11	93.24 ±1.76	83.72 ± 2.41	95.87 ± 2.92	99.85 ± 0.3	99.09 ± 0.84	98 ± 0.95	97.62 ± 1.02
12	95.41 ± 1.38	89.81 ± 1.75	61.83 ± 5.59	87.02 ± 5.65	99.96 ± 0.09	99.74 ± 0.08	99.71 ± 0.31
13	99.96 ± 0.09	100 ± 0	92.71 ± 1.13	99.96 ± 0.09	100 ± 0	99.98 ± 0.04	100 ± 0
OA	93.82 ± 0.15	91.69 ± 0.39	81.36 ± 0.78	91.85 ± 3.39	93.98 ± 1.21	96.92 ± 0.3	**96.94 ± 0.34**
κ	93.12 ± 0.16	90.74 ± 0.44	79.27 ± 0.88	90.91 ± 3.8	93.29 ± 1.35	96.57 ± 0.33	**96.59 ± 0.37**
AA	91.89 ± 0.53	89.87 ± 0.37	79.34 ± 1.2	91.76 ± 2.47	90.99 ± 0.77	**95.13 ± 0.33**	95.09 ± 0.44

and 2.6% average OA, κ, and AA improvements, respectively, in comparison with its old version (i.e., PRCLSTM) for all the experimental datasets, which can verify the effectiveness of the residual learning (applied for PRCLSTM II) in improving the classification outcomes.

D. Analysis of Res-CP Framework

In this section, we first investigate the discrimination power of the Res-CP framework by visualizing the test samples' features in each stage of the classification task. In this regard, t-distributed stochastic neighbor embedding (t-SNE) is used to visualize the samples in a 2-D plot. To do that, we randomly select 200 test samples per class and visualize their features in three stages: 1—raw input data; 2—the features extracted from the last layer of RCN; 3—the features extracted from the last layer of PRCLSTM II. Fig. 12 illustrates the 2-D t-SNE visualization in each of the aforementioned stages for PaviaU, UH, SA, and KSC, respectively. As can be seen from the results, the RCN considerably enhanced interclass dispension

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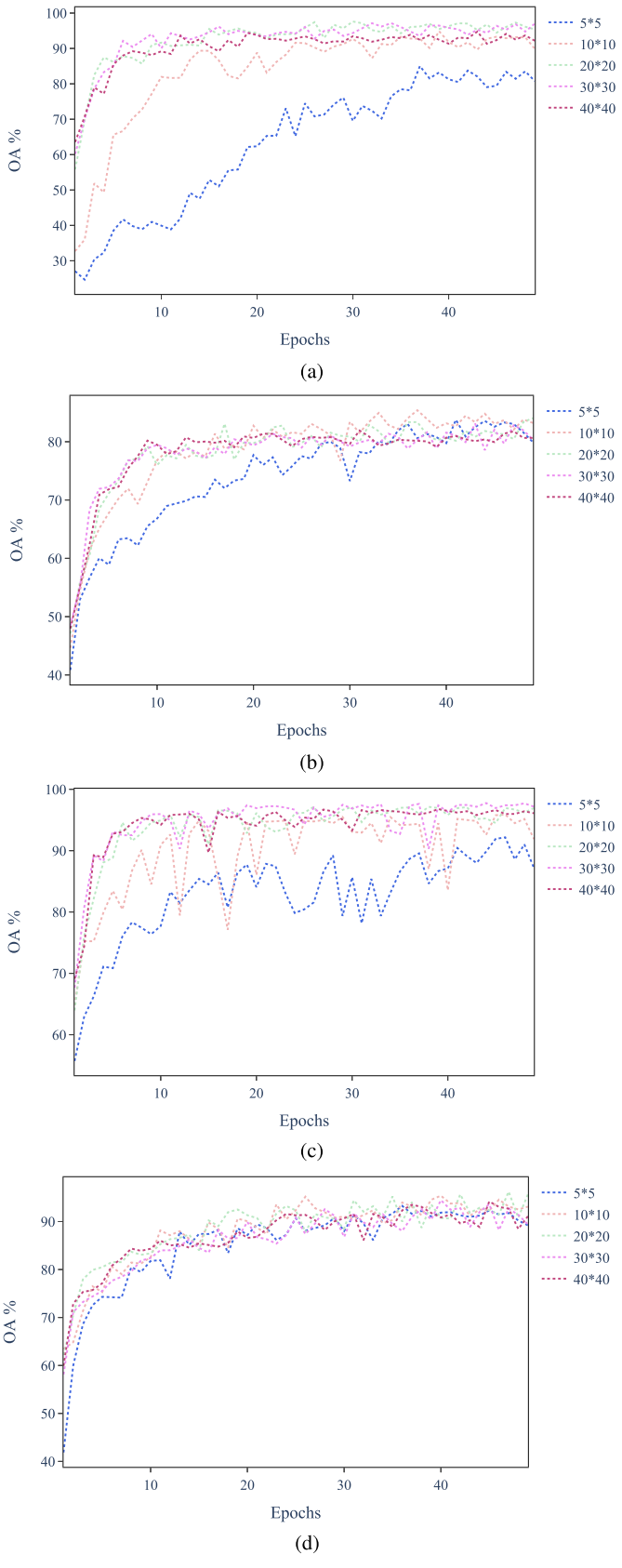


Fig. 11. OA (%) of the Res-CP pseudo-labels in different w' sizes using PRCLSTM II model for datasets. (a) Pavia University (PaviaU). (b) UH. (c) SA. (d) KSC datasets.

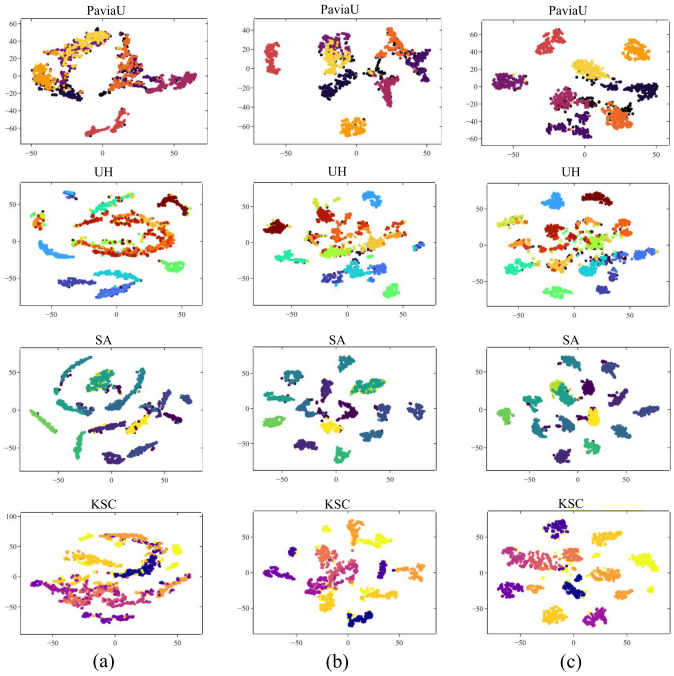


Fig. 12. Embedding features distribution extracted by t-SNE of 200 randomly selected samples of datasets in question. (a) Original image. (b) RCN features. (c) PRCLSTM II features after training with Res-CP pseudo-samples.

and intraclass compactness, thanks to its center-softmax optimization scheme. According to the results of PRCLSTM II, the discrimination of the features is further improved in the next stage; this can imply that the high confidence pseudo-samples selected by the PLG enable the classification model to increase the features and separability, and demonstrate superior classification performance.

Furthermore, we study the influence of PLG hyperparameters on the accuracies by conducting two sets of experiments. First, we investigate the effect of w' in obtaining high-confidence pseudo-samples and final classification outcomes. In this regard, PRCLSTM II is applied to train from scratch using pseudo-samples obtained by considering five different sizes of w' from 5×5 to 40×40. After training in 50 epochs, the PRCLSTM II performance is evaluated on the test set and the OA percentage results are reported. Fig. 11 shows the results of the classification on test sets in 50 epochs for the datasets in question. It is worth noting that the testing assessment is conducted independent of the training task so that the test sets remain totally unseen and PRCLSTM II is prevented from influencing their evaluation results. According to the results shown in Fig. 11, the 5×5 size of w' results in a lower generalization capability of the proposed classification model. This is caused by the fact that the very small size of w' prevents the PLG from utilizing adequate unlabeled samples, and as a result, the generated pseudo-sample sets become small and fail to train the parameters of the classification model well. However, when the size of w' becomes larger, PRCLSTM II converges faster and performs with more stable generalization throughout the training epochs.

Second, we qualitatively evaluate the accuracy of pseudo-sample sets using different sizes of w' by measuring the error rate. The error rate is defined as a ratio of the mislabeled samples divided by the total number of labeled

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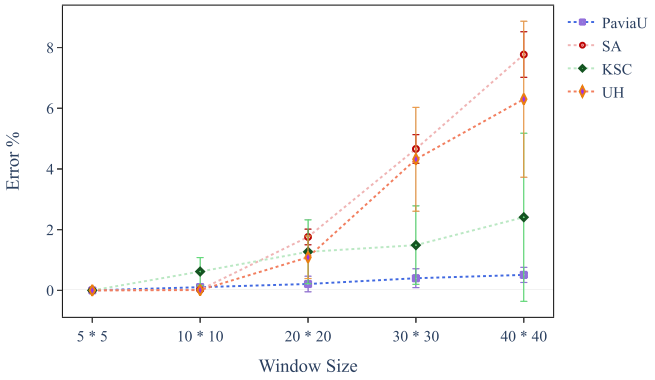


Fig. 13. Error rate (%) analysis of the pseudo-samples generated by Res-CP based on various window sizes (i.e., w') for datasets in question.

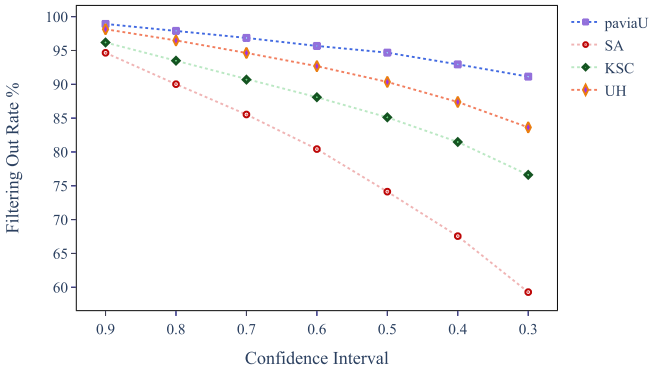


Fig. 14. Filtering out rate analysis of the unlabeled samples during the pseudo-label generation task using different confidence interval (i.e., R^l) level for datasets in question.

samples in the test set. Fig. 13 demonstrates the error rate assessment results of the pseudo-samples obtained with various sizes of w' from 5×5 to 40×40 for the datasets in question. According to the results, the Res-CP is capable of generating pseudo-samples with less than 2% error, that is, 98% correct labels, when the size w' is 20×20 or smaller. The error rate is increased in large w' sizes. However, as can be seen from Fig. 11, the rate of increase is not substantial such that it degrades the generalization performance of the classification model. This is caused by the fact that the number of correct pseudo-samples considerably outweighs the number of incorrect ones.

Third, we quantitatively analyze the effect of R^l in filtering out unconfident unlabeled samples during the pseudo-label generation task. To this end, a set of experiments with different levels of R^l from 0.3 to 0.9 are performed to analyze the rate of unlabeled samples that are discarded during the PLG pseudo-label generation. Fig. 14 shows the results of the filtering out analysis with the different levels of R^l for the experimental datasets. According to the results, fewer samples are discarded by relaxing the R^l level. However, a large degree of discrimination still exists among the estimated labeled distributions and the unlabeled samples. This is caused by the fact that the center-based embedding task, performed by the RCN, leads to a large degree of discrimination among labeled and unlabeled samples.

Finally, we assess the computational burden of the RCN and PLG components of the Res-CP framework for generating the pseudo-labels. According to Table X, the computational time

TABLE X
COMPUTATIONAL TIME (MIN: MINUTES, S: SECOND) OF THE RES-CP COMPONENTS FOR THE FOUR DATASETS

Model	Time	PaviaU	UH	SA	KSC
RCN	Train	0.87 m	1.75 m	2.27 m	1.3 m
	Inference	0.88 m	4.6 m	0.8 m	1.4 m
PLG	Train	-	-	-	-
	Inference	8.5 s	5.9 s	3.7 s	2.5 s

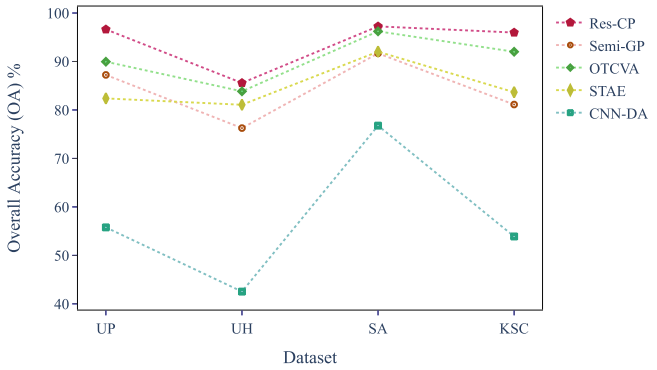


Fig. 15. Classification accuracy (%) of the Res-CP and its rivals on all the experimental datasets.

of the two components of the Res-CP framework is reasonable, ranging from 2.5 s to 4.6 min, for the experimental datasets.

E. Assessment of Res-CP

In this section, we assess the performance of the proposed semisupervised learning (SSL)-based Res-CP against its counterparts. To this end, five comparative methods with different approaches are applied to address the limited labeled data problem in HSI classification. The competing methods, including CNN-DA [82], orthogonal total variation component analysis (OTVCA) [83], [84], self taught-autoencoder (ST-AE) [85], and SSL-graph prototypical (GP) [55], [56], are briefly explained in the following.

1) CNN-DA: A DA-based method, which generates a set of occluded images as the training samples to enhance the performance of the CNN-based classification method.
2) OTVCA: A two-stage unsupervised feature extraction and classification method. Specifically, the extracted features are consumed by the RF classification method.
3) STAE: A classification method with a combination of self-supervised and supervised learning approaches to address the problem of limited labeled data of HSI.
4) Semi-GP: A SSL-based method that applied a graph-based DNN to perform HSI classification with a limited number of training samples.

All the competing methods are configured with their optimal hyperparameter settings and trained using the same randomly selected training and test sets. The assessment results based on OA % accuracy are demonstrated in Fig. 15. According to the results of Fig. 15, the proposed framework (i.e., Res-CP) consistently outperformed its rivals and yielded superior OA test accuracy in all the experimental datasets.

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V. Conclusion and Remarks

In this article, we proposed a deep probabilistic framework to generate reliable pseudo-labels for HSI semisupervised classification. Unlike the existing deep semisupervised methods, our proposed framework does not require a fine-tuning stage and also scales well to the large-scale HSI dataset (i.e., UH); improving up to 26.63% in OA of the classification results. We show that by representing labeled samples as a Gaussian distribution in the latent space, the similarities among labeled and unlabeled samples can be effectively captured. In addition, the properties of the distribution enable us to filter out noisy and low-confidence unlabeled samples. To our knowledge, this is the first work that applies probabilistic embedding for HSI pseudo-label generation. Extensive experiments demonstrate significant and consistent improvement in the performance of several DNNs by explicitly learning from pseudo-samples. In the future, we will investigate the application of the proposed framework for other types of image data (e.g., medical images).

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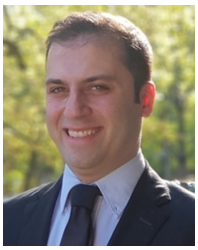
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